



April 12, 2025

Cover Letter – Resubmission of PONE-D-25-00468

Title: *Is There a Competitive Advantage to Using Multivariate Statistical or Machine Learning Methods Over the Bross Formula in the hdPS Framework for Bias and Variance Estimation?*

Dear Dr. Adineh,

We thank you and the reviewers for the thorough and constructive feedback on our manuscript (PONE-D-25-00468). We are pleased to submit a revised version of our article entitled “Is There a Competitive Advantage to Using Multivariate Statistical or Machine Learning Methods Over the Bross Formula in the hdPS Framework for Bias and Variance Estimation?” for reconsideration in *PLOS ONE*.

We have carefully addressed all comments from the reviewers and the journal’s editorial team. A detailed Response to Reviewers is included, outlining how each point has been addressed. Revisions have been made in both the main text and figures/tables, with changes marked in red in the tracked version of the manuscript.

Below, we address the journal’s additional requirements:

1. Manuscript formatting:
We have ensured that the manuscript complies with *PLOS ONE* formatting requirements, including file naming and style.
2. Code availability:
All author-generated code and analysis scripts are openly available at our GitHub repository: <https://github.com/ehsanx/hdPS-proxy-select-codes>.
A citable, stable version of the code is archived on Zenodo:
<https://doi.org/10.5281/zenodo.15201233>.
3. Financial disclosure – Role of funders:
This work was supported by MEK’s NSERC Discovery Grant (PG#: 20R01603) and Discovery Launch Supplement (PG#: 20R12709).
The funders had no role in study design, data collection and analysis, decision to publish, or preparation of the manuscript.
4. Competing interests statement:
MEK is currently supported by grants from CIHR and MS Canada. MEK has previously received consulting fees from Biogen Inc. for consulting unrelated to this current work. MEK was also previously supported by the Michael Smith Foundation for Health Research Scholar award.
This does not alter our adherence to PLOS ONE policies on sharing data and materials.
5. Data availability:
The minimal dataset and supporting files required to replicate our findings are deposited



in a stable public repository on Zenodo:
DOI: 10.5281/zenodo.15201233

6. Ethics statement:

The ethics statement has been relocated to the Methods section, as requested. The study involves only secondary de-identified data and complies with the ethical guidelines outlined by UBC's Policy 89 and TCPS2.

We sincerely hope that the revisions we have made adequately address the feedback and improve the clarity, rigor, and relevance of our work. Thank you for the opportunity to revise our manuscript.

Please do not hesitate to contact us if further clarifications are needed.

Sincerely,

A handwritten signature in blue ink that reads 'M Ehsan Karim'.

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